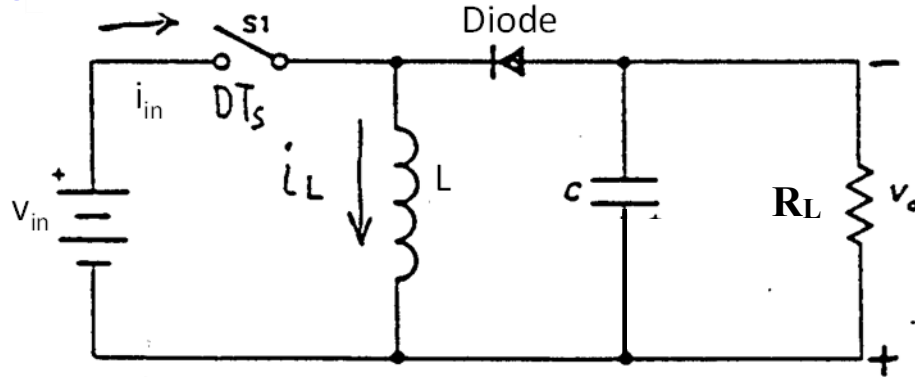


ECE 5254 - Homework #2

Three-terminal PWM switch model

1. Analysis of Buck-Boost converter (*Diode and Switch are both ideal*)
using three-terminal PWM switch model:



With circuit parameters as follows: $V_{in}=12\text{ V}$, $V_o=4\text{V}$, $C= 300\text{ uF}$, $F_{sw}= 300\text{ kHz}$, $R_L= 2\text{ }\Omega$.

1-A. Continuous Conduction Mode (CCM) Case [$L_s = 10\mu\text{H}$] (25%)

- Derive DC Model. (5%)
- Derive Small-signal equivalent circuit model. (5%)
- Derive transfer function of control to output $G_{vd}(s)=v_o(s)/d(s)$, and output impedance $Z_o(s)= v_o(s)/i_o(s)$. (10%)
- Verify the model results of $G_{vd}(s)$ and $Z_o(s)$ with SIMPLIS simulation. (5%)

1-B. Discontinuous Conduction Mode (DCM) Case [$L_s = 0.5\mu\text{H}$] (35%)

- Derive DC Model. (7.5%)
- Derive Small-signal equivalent circuit model. (7.5%)
- Derive transfer function of control to output $G_{vd}(s)=v_o(s)/d(s)$, and output impedance $Z_o(s)= v_o(s)/i_o(s)$. (15%)
- Verify the model results of $G_{vd}(s)$ and $Z_o(s)$ with SIMPLIS simulation. (5%)

Notes:

The transfer function should be written in the analytical polynomial form as follows to get full points:

$$G_{vd}(s) = G_o \frac{(1 - s/\omega_{z1})(1 + s/\omega_{z2})}{1 + s/Q_o\omega_o + s^2/\omega_o^2}$$

You need to use symbols of the circuit components, e.g. “ L ”, “ R_L ”, and derive an “analytical expression” for the control-to-output and output impedance transfer functions. Please show all necessary steps. The circuit parameters given above are for simulation verification. Numerical transfer function should not be provided for questions 1-A(iii) and 1-B(iii). In questions 1-A(iv) and 1-B(iv), you must plot the model result and SIMPLIS simulation result in the same bode plot for each transfer function to see whether the two results can match with each other. The start and stop frequencies should be 100 Hz and 500 kHz, respectively.

1-C. Case Discussion (Your discussion should be rich enough.) (20%)

- (i) What are the differences you observe between the control-to-output response and output impedance characteristics in CCM and DCM conditions? Will DCM reduce converter transfer function order? Does the system exhibit first order or second-order response in DCM?
- (ii) Does the magnitude plot of DCM control-to-output transfer function contain resonant peaks? If not, mathematically and physically explain the origins of additional damping using the transfer function and three-terminal model you derived in question 1B.
- (iii) Will DCM eliminate the right-half plane zero (RHPZ) in buck-boost converters? Explain the reasons for your stance.
- (iv) Analytically compare the relative locations of RHPZ in CCM and DCM conditions? which operation mode move RHPZ to higher frequency CCM or DCM?

2. General Discussion. (20%) (Your discussion should be rich enough.)

You must discuss the following:

- (i) To derive the DCM three-terminal switch model, the i_{pk} (peak current of inductor) is used. However, it is not used while deriving the CCM model. Does it mean the switching ripple is considered in small-signal model under DCM operation? State your opinion and explain the reason.

ii) For the small-signal model of PWM switching converters based on three-terminal PWM switch concept, will f_{sw} (switching frequency) influence small-signal model? State your opinion and explain the reason. (Consider operation of CCM and DCM)

Due time: Oct. 17 (Thursday), 2024. 11:30pm. Only .pdf file is accepted. **It is preferred to type your solutions.** If you must write, please make sure it is clear and easy to recognize, also highlight your final results. **If it cannot be recognized, you will not get the points.** Please change your file name to HW#_First name _Last name for easy recognition. (For example, HW2_Sundaramoorthy_Sridhar.pdf).

All the homework should be submitted into canvas before due time. Late homework should be sent into TA's email sundaramoorthy@vt.edu (10% penalty per day after due time will be strictly executed unless prior arrangements have been made with the instructor).

All the work MUST be your own; the VT honor code will be strictly enforced.